EPSILON SOUNDS

USER MANUAL · VERSION 1.2

Epsilon Bounce

Bounce takes any audio loop (drums, a melody, a flipped sample), cuts it where the sound changes, and plays the pieces back on your project's tempo in the pocket you choose: dembow, drill, amapiano, boom bap and nine more, or the exact feel of your own drums, so your melody finally sits in the same groove as your beat.



Locks melodies to your drums · **Drums · Melodic · Grid cutting** · **13 pockets** · **42 presets** · **Half / double time** · **VST3 · AU · Standalone** · **0.1 % CPU on an Intel Mac**

WHAT IT DOES

Four steps, no time-stretching

Bounce never stretches audio. It cuts the loop at each hit and fires each piece at a new moment, so drums stay as punchy as the original at any tempo.

LOAD

Drop a WAV, AIFF or FLAC onto the grid, or press `LOAD LOOP`. Up to 60 seconds: drums, percs, melodies, vocal chops.

CUT

On drum hits, on note and chord changes for melodies, or on an even beat grid for dense samples. Auto-align fixes loops that start a few ms off.

RE-GROOVE

Each slice is pulled onto the 16th grid by **TIGHTEN**, then moved by the pocket: some hits late, some early, some louder.

PLAY & DRAG

Slices fire from FL's playhead, sample-accurate. `DRAG OUT` drops the result into a track as a 24-bit WAV.

THE BIG ONE

Lock your melody to your drums

A melody or sample that "doesn't fall" with your drums is almost always one of four problems. Bounce fixes each one separately, so you can hear which it was.

A It swings differently

Your drums lean one way and the sample another, so notes land between the drum hits. **Match drums** learns your drum loop's exact pocket; the sample's notes move to it. Your drum hits appear as markers on the grid so you can see the notes lock under them.



B It's early or late as a whole

Many samples are cut a few milliseconds before or after the beat, or start on a pickup. **Auto-align** finds where the sample's beat really sits (it shows "Aligned +12 ms" on the grid). **Shift** slides bar 1 by whole 16ths for pickups.



C There are no hits to cut

Pads, chords and vocals change notes without a drum-like attack, so a transient detector sees nothing. **Cut on: Melodic** listens to which notes are sounding and cuts where the harmony changes. **Cut on: Grid** chops dense samples every 16th, 8th or beat, on zero crossings so nothing clicks.



D Moving notes leaves holes

Push a chord later and there's silence where it used to start. **Hold** keeps each note sounding, using a seamless loop of its own tail, until the next one arrives. Cuts crossfade over 8 ms. Hold is automatic for Melodic and Grid.



The 30-second lock

1. Load your melody or sample with `LOAD LOOP`.

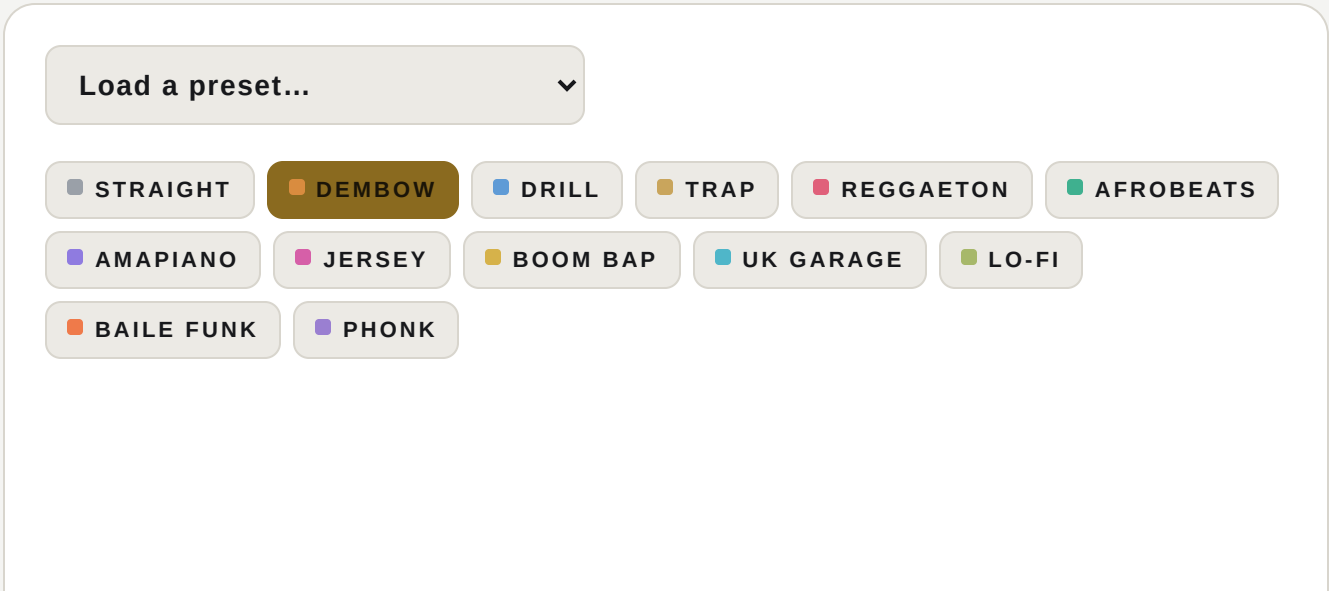
2. Set **Cut on** to **Melodic** for chords, keys, guitar or vocals, or **Grid** for a dense flip (8TH for soul samples, 16TH for busy ones).
3. Bounce your drum loop to audio in FL and press `MATCH DRUMS`, or hold `ALT` / `OPTION` and drop it on the grid. Bounce switches to LEARNED.
4. Raise **Tighten** to 80–100 % so the sample's own wobble is cleaned up before your drums' feel goes on.
5. If the downbeat still sounds off, try **Shift** ±1 or ±2 (pickups), and check **Bars** is the sample's real length.
6. Set **Accent** to 0 so chord volumes stay even, then press play with your drums.

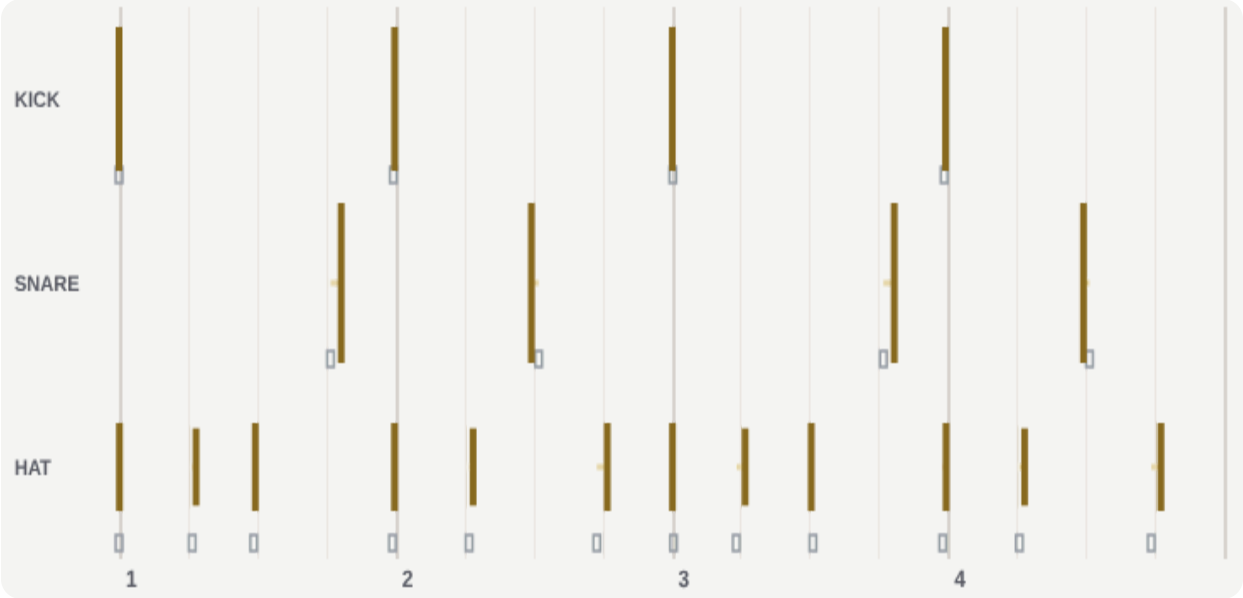
Try it in the Pocket Lab below: switch it to **Melody lock** to hear a chord loop played straight against grooved drums, then locked to them.

HEAR IT

Pocket Lab

This is Bounce's own groove maths running in the page. Pick a pocket or a preset and watch where each hit moves. In **Drums A/B**, play alternates the loop straight and grooved, one bar each. In **Melody lock**, the drums are always grooved and a chord loop alternates between straight (the "doesn't line up" sound) and locked to the drums, with Hold.





where the hit was where the pocket puts it the move

Amount	Swing	Tighten	Humanize
100 %	0 %	60 %	2.0 ms
Accent	**Feel**	**Time**	**Tempo**
75 %	0.0 ms	Normal	100 BPM

The demo loop is played a little loose on purpose, the way a real sample is, so you can hear Tighten clean it up.

THE PANEL

Where everything is



1. **Load loop**: pick the loop to re-groove. Dropping a file on the grid does the same.
2. **Preset browser**: 42 presets grouped by genre. The arrows step through them; presets never touch your loop, cutting or output level.
3. **Preview / Drag out**: Preview plays with FL stopped, at the Tempo knob. Hold Drag out and drop into a track for a WAV of two loops.
4. **Cut on**: DRUMS cuts on hits, MELODIC on note and chord changes, GRID on an even beat grid. Switching re-cuts the loop instantly.
5. **Your drums**: after Match drums, each marker is where your drum loop hits. The notes below line up with them.
6. **Groove Grid**: hollow ticks show where each piece was; champagne bars show where the pocket puts it; the playing piece glows. "Aligned" shows any auto-align correction.
7. **Pocket chips**: the 13 genres plus LEARNED, the feel copied from your drums.
8. **Sensitivity, Shift, Tempo**: how many cuts are found; slide bar 1 by whole 16ths; the Preview speed when FL isn't playing.
9. **Bars / Host sync**: override the loop's length in bars; follow FL's tempo and transport.
10. **Grid chop / Hold**: 16th, 8th or beat chops in Grid mode; Hold cycles AUTO, ON and OFF.

11 **Match drums / Auto-align**: learn your drum loop's pocket; fix loops cut a few ms early or late.

12 **Feel**: Amount is the hero: how much of the pocket. Swing, Tighten, Humanize, Accent, Feel and Output shape it.

13 **Time**: Normal, Half-time or Double-time playback of the loop's grid.

EVERY CONTROL

What each knob does to the hits

Each diagram is one bar of 16ths: grey ticks are the loop as it came in, champagne is the result.

Amount

0 – 150 %, default 100 %

How much of the pocket is applied. 0 % leaves only Tighten and tempo-follow; over 100 % exaggerates the genre past itself.



Swing

0 – 100 %, default 0 %

Extra delay on every off-16th, on top of the pocket's own. 100 % is a full triplet feel (75 % MPC swing).



Tighten

0 – 100 %, default 50 %

Pulls the loop's own timing onto the 16th grid before the pocket moves it. 0 % keeps the sample's feel, 100 % is fully quantized.



Humanize

0 – 20 ms, default 0 ms

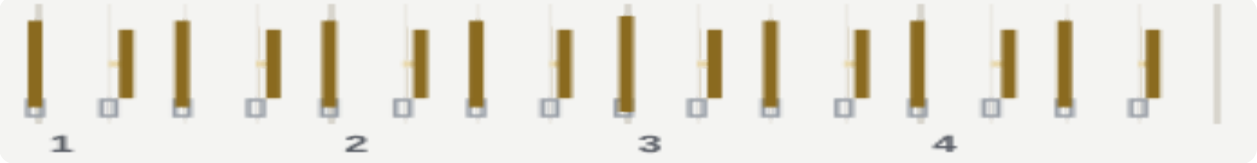
A small random move per hit, the same on every playback, so prints and takes always match.



Accent

0 – 100 %, default 75 %

Applies the pocket's louder and softer steps. 0 % moves only timing, which suits melodic loops.



Feel

±20 ms, default 0

Moves the whole loop ahead of the beat (−, urgent) or behind it (+, lazy). The first hit never moves before the bar.



Cut on

Drums, Melodic, Grid

What Bounce cuts on. Drums: sudden hits. Melodic: changes in which notes are sounding, so smooth chord changes and legato lines cut cleanly and vibrato doesn't split a note. Grid: even chops on the beat for dense samples.

Sensitivity

0 – 100 %, default 50 %

How many cuts are found. Drums: higher catches ghost notes; lower keeps rolls glued to their main hit. Melodic: higher splits arpeggios into single notes; lower cuts only on chord changes.

Grid chop

16th, 8th, Beat

In Grid mode, how often the sample is chopped. 8th is the classic soul-flip size; 16th for busy samples; Beat for slow pads.

Hold

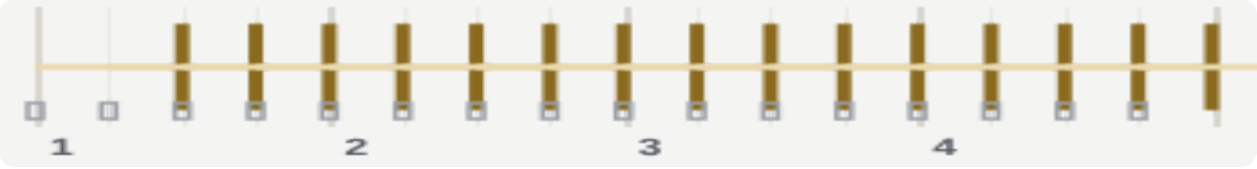
Auto, On, Off

Keeps each piece sounding until the next one starts, using a seamless loop of its own last 20–80 ms, so moved notes and half-time leave no gaps. Auto: on for Melodic and Grid, off for Drums (tight drum gaps are part of the sound).

Shift

−8 to +8 sixteenths

Slides where bar 1 starts. Use it when a sample begins on a pickup or the chord change you hear as 'one' isn't at the start of the file.



Auto-align

On / Off, default on

Measures where the loop's loudest hits sit against the grid and corrects samples cut a few ms early or late (anything over 1.5 ms). The correction shows on the grid.

Bars

Auto, 1, 2, 4, 8

The loop's length in bars, which sets the grid. Auto picks the length that puts it between 85 and 170 BPM.

Time

Normal, Half-time, Double-time

Plays the loop's grid at half or double speed. See Time modality below.

Output

−24 to +6 dB

Output level. Presets never change it.

THE POCKETS

13 grooves, 42 presets

Every pocket is a map of the 16 steps in a bar: how early or late each one sits, plus whether it's louder or softer. The bar under each step shows the move: up is late, down is early, and longer means further. Each preset is a different take on the pocket: more swing, tighter, pushed or laid back, or played at half or double time.

STRAIGHT

swing 50 %

No pocket: just Tighten and tempo-follow. The clean base for fixing loops.

Quantize Clean
Snaps every hit to the grid and follows your tempo - a slice-based time-stretch with no smearing.

Human Drummer
Keeps 40 % of the original timing and adds a few ms of drift that is the same every playback.

Double-Time Roll DOUBLE
Plays the loop twice as fast on the grid - instant rolls and fills from any drum loop.

DEMBOW

swing 55 %



Kick on every beat; the 'ch' on 16ths 3, 6, 11, 14. The 3s lean late and the 6s push, with light swing.

Dembow Clasico
The textbook pocket: the 'ch' on 3 and 11 leans late, 6 and 14 push.

Dembow Romantico
Softer, laid back 6 ms, a little shuffle - for the slow, melodic side.

Dembow Callejero
Exaggerated lean, hard accents, pushed ahead of the beat - street energy.

Dembow Half-Time Perreo HALF
Half-time: the loop stretches over twice the bars for a heavy, slow perreo.

DRILL

swing 62 %



Heavy 16th shuffle (~62 %), the half-time snare dragging a hair, the 16th before it pulled in.

Drill UK Classic
The ~62 % shuffle with the dragging half-time snare.

Drill NY Bounce
More swing and pushed forward - the Brooklyn bounce.

Drill Bronx Slide
Heaviest swing, laid back - hats slide into the snare.

Drill Half-Time Slide HALF
Half-time drill: a fast loop becomes the slow, sliding drill frame.

TRAP

swing 52 %



Straight and tight; downbeats a touch early, off-8ths quieter.

Trap Atlanta Tight
Dead tight, downbeats a hair early - classic Atlanta.

Trap Melodic Float
Loose and floating for melodic trap and R&B.

Trap Rage Rush
Everything pushed 8 ms early, accents slammed - rage urgency.

Trap Hi-Hat Double DOUBLE
Double-time: a hat or perc loop becomes 32nd-note rolls.

REGGAETON

swing 50 %



Dembow's cleaner cousin: the same snare pattern almost on the grid, accented harder.

Reggaeton Comercial
Clean radio dembow: almost on the grid, accents doing the work.

Reggaeton Old School
2000s feel: more lean, more swing, laid back.

Moombahton Half-Time HALF
Half-time reggaeton - the moombah tempo from a reggaeton loop.

AFROBEATS

swing 58 %



A rolling 3-3-2 on 0, 3, 6 and 8, 11, 14, with light swing; the 3s lay back.

Afrobeats Lagos Roll
The rolling 3-3-2 with light swing.

Afro-Swing
London afro-swing: more shuffle, laid back.

Afro-Drill
Afro accents with drill-level swing, pushed - the crossover.

AMAPIANO

swing 60 %



Shuffled shakers (~60 %) and log-drum off-8ths laid back; the kick dead on.

Amapiano Log Drum Lazy
Shakers shuffled, log-drum 8ths late - the yanos pocket.

Amapiano Private School
Soft, jazzy and very late - private-school piano.

Amapiano Bacardi
Tighter and punchier for the Bacardi / gqom edge.

JERSEY

swing 52 %



The five-kick bounce (0, 4, 8, 10, 12) pushed early, everything else ducked.

Jersey Bed-Squeak Bounce
The five-kick bounce, pushed and ducked around the kicks.

Philly Club
A little swing on top - the Philly cousin.

Jersey Double-Time Run DOUBLE
Double-time: half-speed loops become 140+ club runs.

BOOM BAP

swing 58 %



MPC 58 % swing with the backbeat snare laid back like a live drummer.

Boom Bap Golden Era
MPC 58 % with the snare sitting back.

Boom Bap SP-1200 Dusty
Looser, dustier, a touch more swing.

Boom Bap Dilla Drunk
Barely quantized, exaggerated swing and drift - the drunk feel.

UK GARAGE

swing 66 %



2-step: a 66 % shuffle, the skipping kick on 10 early, snares a hair late.

UKG 2-Step Skip
The 66 % shuffle with the skipping kick.

Speed Garage
Tighter and pushed for the speed garage drive.

UKG Bassline Swing
Extra swing for bassline / niche.

LO-FI

swing 57 %



Drunk and late: everything but the downbeat drags, accents flattened.

Lo-Fi Study Session
Lazy and late, accents flattened.

Lo-Fi Tape Wobble
Loose timing and 12 ms of drift - a worn tape machine.

Lo-Fi Half-Time Rain HALF
Half-time and late - for slowed, rainy beats.

BAILE FUNK

swing 53 %



Tamborzão on 0, 3, 6, 10, 12: the 3 and 10 rush, the 6 sits back.

Baile Funk Tamborzao
The tamborzao: the 3 and 10 rush, the 6 sits back.

Funk Mandelao
Exaggerated and slammed - mandelao aggression.

Funk BH
Looser, swung, laid back - Belo Horizonte style.

PHONK

swing 56 %



Memphis bounce: medium swing, the half-time snare heavy and late.

Phonk Memphis Cowbell
Medium swing, the half-time snare heavy and late.

Drift Phonk
Tighter, pushed, accents slammed for drift phonk.

Phonk Half-Time Haunt HALF
Half-time and late - the haunted Memphis crawl.

TIME MODALITY

Half-time and double-time

Half-time plays the loop's grid at half speed: a 2-bar loop now fills 4 bars of your project, every gap twice as long. Use it to turn a 140 BPM drill loop into a 70 BPM half-time slide, or reggaeton into moombahton.

Double-time plays it twice as fast: a loop's 16ths become 32nds. A hat or perc loop turns into rolls; a slow Jersey loop becomes a club run. The pocket still applies, so double-time trap hats keep their swing.

Because Bounce fires slices instead of stretching audio, each hit keeps its original pitch and punch. Half-time leaves short gaps after each hit, which is the half-time sound. Drag-out file names say which mode was used.

MATCH DRUMS

Match Drums: copy the feel of anything

Press MATCH DRUMS and pick your drum loop (or any reference), or hold ALT / OPTION while dropping it on the grid. Bounce listens to it, measures how far every hit sits from the grid on each of the 16 steps, and how loud each step is. That becomes the LEARNED pocket, applied to your loop with every Feel knob still working.

The grid then shows your drum hits as markers along the top, with the name and the swing it heard as an MPC-style percentage. Steps it never heard borrow the timing of steps with the same role (downbeats, 8ths or 16ths), so a hat pattern's swing still carries over to a loop with more hits. The learned feel is saved inside your FL project.

Learning uses a slightly higher sensitivity than slicing, because ghost notes carry most of a groove's feel. If the reference reads at the wrong length (the swing figure looks odd), set **Bars** to the reference's real length before learning.

TIPS AND TRICKS

Things nobody tells you

Each of these follows from how Bounce actually works inside, and none is obvious from the panel.

Flip a soul sample in one move

Grid, 8TH chops, Hold on, then Boom Bap Golden Era. Every 8th of the sample swings like an MPC flip and the chords sustain through the swing, without chopping anything by hand.

Cut on Grid · 8TH · Boom Bap

Turn chords into a dembow

Melodic cutting on a pad or piano loop, Sensitivity low so each piece is a whole chord, then a Dembow preset with Hold off. The chord stabs bounce on the ch-ch with short, rhythmic gaps between them.

Melodic · Sensitivity 25 % · Hold off · Dembow

Slowed without the pitch drop

Half-time with Hold on gives a slowed, spacious version of a melody that keeps its original key, unlike slowing the sample down. Pair it with Lo-Fi Half-Time Rain for the slowed-and-reverb feel without detuning.

Half-time · Hold on · Lo-Fi

Find the real "one"

If a sample's loop point isn't its downbeat, step Shift through −4…+4 while it plays with your drums. The setting where the chord changes land on your kick is the one.

Shift while playing

Clean tempo changes, no artefacts

Pick Straight, Tighten 100, Amount 0. Bounce now only follows tempo, slice by slice, so a 90 BPM loop drops into a 140 session without smearing or pitch change, which a time-stretch can't promise.

Preset: Straight · Quantize Clean

Keep a roll in one piece

A slice carries everything until the next detected hit. Lower **Sensitivity** and a hat roll or flam stays glued to the hit before it and moves as one, instead of each tick being re-grooved separately.

Sensitivity 20–35 %

Swing 32nds or 8ths with Bars

Bars tells Bounce how long the loop is, and that sets the grid. Tell it a 2-bar loop is 4 bars and the "16ths" become 32nds, so the swing lands on 32nd hats. Tell it 1 bar and the swing moves to 8ths: instant boom-bap 8th shuffle.

Bars: double or half the real length

Double-tracked width

Humanize uses the same random pattern every time and scales it with the amount. Put the loop on two channels, pan them hard left and right, and set Humanize to 3 ms on one and 9 ms on the other: the hits flam differently on each side, and it sounds like a live double.

Two instances · Humanize 3 / 9 ms

Pushed hats, lazy snare

Split the loop into two copies, high-passed hats and low-passed kick/snare. Bounce both with the same pocket, then set Feel −6 ms on the hats and +8 ms on the drums. The classic "hats ahead, snare behind" tension comes from one loop.

Feel −6 / +8 ms on two instances

Put a melody in a drummer's pocket

Learn the groove from a drum break you love, then load a melody or chord loop. Every note change is a slice, so the chords now sit exactly where that drummer played. With Sensitivity low, each chord stab moves as one piece.

Learn a break → load the melody

Your own feel on every loop

Finger-drum 2 bars of hats on your pads, bounce that to audio, and Learn it. Now every loop in the beat (percs, guitar, vocal chops) can share your timing, so the whole track grooves like one player.

Record 2 bars → Learn → LEARNED on every track

Stutter gaps on vocal chops

Moved-late hits end in a 4 ms fade, so they leave clean silence instead of a click. Push Feel to +15 ms with Amount 150 % on a vocal chop loop and you get tight, rhythmic gaps that sound edited by hand.

Feel +15 ms · Amount 150 %

Make new samples to sell or flip

Drag out renders at the project tempo with the current pocket and Time mode. Change the session BPM, pocket or mode and drag again: one loop becomes a pack of tempo- and genre-matched variations, ready for Epsilon Chop.

Change BPM / preset → Drag out → repeat

Audition with FL stopped

Turn on `PREVIEW` and step through presets with the arrows while FL is stopped. Set the Tempo knob to your project BPM so what you hear matches what you'll get on play.

Preview + preset arrows

Exaggerate, then flatten

Amount over 100 % pushes a pocket past the genre. Pair it with Accent 0 to move the timing without changing levels, which is ideal for melodic loops where volume jumps would sound wrong.

Amount 130–150 % · Accent 0

Fix a sloppy sample before anything else

Old vinyl breaks drift. Straight with Tighten 100 snaps them to the grid; then pick Boom Bap and the swing is applied to clean timing instead of piling on top of the drift.

Tighten 100 first, then a pocket

UNDER THE HOOD

How Bounce hears and plays

Finding hits

A spectral-flux detector compares each 6 ms frame with the last and flags sudden new energy. An adaptive threshold set by Sensitivity ignores tails and reverb; hits closer than 45 ms merge.

The real attack

Each hit is traced back sample by sample to where the sound starts rising, then 3 ms of pre-roll is kept. Bounce fires each slice early by that pre-roll so the attack itself lands on the target.

Hearing notes

Melodic mode builds a 12-note map of each moment from the spectrum's peaks, with each frequency refined between FFT bins. It compares the notes in the next 120 ms with the last 120 ms; a big difference is a chord or note change. The short-window detector then pins the exact moment the new notes enter.

Hold

When a piece ends before the next one starts, its last 20–80 ms loops with a cosine crossfade into matching earlier samples, so the note keeps its pitch and tone. Holds stop after 4 seconds with a 50 ms fade.

Auto-align

The loudest hits (kicks and snares, which are rarely swung) vote on how far the loop sits from the 16th grid. Their weighted median is applied only when they agree, so a swung loop isn't mistaken for a late one.

Bar count

Auto picks the number of bars that puts the loop between 85 and 170 BPM, where trap, drill and dembow live. Override it with Bars when a loop is unusual.

Placement

For each hit: start from where it was, pull it toward the nearest 16th by Tighten, add the pocket's offset for that step times Amount, the Swing knob on off-16ths, then Feel and Humanize in milliseconds.

Playback

Slices trigger from FL's playhead with sample accuracy; tempo changes and loop jumps re-sync instantly. A new hit chokes the last with a 4 ms fade. No audio is allocated while playing.

Light

Measured on GitHub's Macs: about 0.12 % of one core on an Intel Mac, 0.02 % on Apple Silicon. Slicing runs on a background thread so loading never glitches playback.

FIXES & FAQ

If something sounds off

▼ **My melody still doesn't sit with the drums**

Work through the four fixes in order: Match drums (feel), Auto-align on and Shift (offset), Cut on Melodic or Grid (cuts), Hold on (gaps). Also check Bars: a 4-bar sample read as 2 bars will never line up.

▶ **Notes get cut off or sound choppy**

▶ **The loop plays at double or half speed**

▶ **Hits sound doubled or flammy**

▶ **Ghost notes don't move with the groove**

▶ **Nothing plays**

▶ **The groove feels too much**

▶ **Learn says "not enough hits"**

▶ **macOS says the plugin is damaged**